BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Spring Assessment (AP2)

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is a typical feature of proprietary software?

2. Why use a different password for each account?

A) Make all passwords shorter
B) Remove the need for login
C) Limit damage after a leak
D) Give everyone equal access

3. "Defragmentation must speed up every SSD by moving its read head." Correct this. Explain why.

4. A program saves results and later reads attendance data. Correct both claims. Explain why. (a) Closing a file deletes it. (b) A program can read a file without opening it.

Today's objectives

Recall key knowledge independently before the assessment. Show accurate working on calculations, pseudocode and explanations.

Key vocabulary - Read, cover, recall

Method marks
Credit for correct working even if a final numerical result is wrong.

Trace
A record of the executed operations and changing variable values.

Command word
An instruction such as state, explain, calculate or evaluate.

Assessment
An independent measure of current knowledge and skills.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Read the question	Underline the command word and identify the information provided.
Allocate the marks	For explain and compare questions, include distinct relevant causal points.
Show your working	Give calculation steps with units; record indexes, tests and intermediate values.
Check answers	Review the final value, boundaries, condition logic and terminology.

I Do - One worked example

A separate revision example: calculate the uncompressed size of a 10 x 10 bitmap with 8-bit colour depth, in bytes.

1. Read the width 10 pixels and height 10 pixels.
2. Find 10 x 10 = 100 pixels.
3. Multiply 100 x 8 = 800 bits.
4. Convert 800 / 8 = 100 bytes. This example is not from AP2.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Trace total=0; for i=1 to 3; total=total+i; next i. State each total and the printed value.

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the output. Then explain the variable's change.

```
total = 0
for i = 1 to 3
total = total + i
next i
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. Explain why showing steps can earn marks even if a final calculation is wrong. [2]

Q3. Explain why an answer should use the question's command word. [2]

Q4. State one reason to check the output of a trace table. [1]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Show calculation steps and state units where a question needs them. ☐ T ☐ F
2. Underline the command word and identify the information provided. ☐ T ☐ F
3. For explain and compare questions, include distinct relevant causal points. ☐ T ☐ F

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Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Source code is normally controlled by its owner and users receive specified usage rights. ☐

Q2 C - Limit damage after a leak ☐

Q3 It brings file fragments together, reducing read-head movement when accessing a file. ☐

Q4 (a) Closing releases the file handle and helps finish buffered writes; it does not delete the file. (b) Opening creates a handle for access. Open, read or write, then close. Screen output does not save a file. ☐

Code First - Model

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We Do - Check the model

1. After i=3 total becomes 6. Output: 6. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Correct method can earn method credit [1]; a visible step enables identification of what was done accurately [1]. ☐

Q3 State/name and explain/evaluate require different levels of detail [1]; matching the command focuses on assessable evidence [1]. ☐

Q4 Checking that the final output agrees with executed variable updates [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: